

*Total No. of Questions : 8]*

*[Total No. of Printed Pages : 2*

**Roll No .....**

**MI-6002-CBGS**

**B.E. VI Semester**

Examination, June 2020

**Choice Based Grading System (CBGS)**

**Mining Environment - II**

***Time : Three Hours***

***Maximum Marks : 70***

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) What is theoretical head characteristic of a Centrifugal fans? Explain the same for all types of centrifugal fans.  
b) What are the various Efficiencies of mine fans? Explain them.
2. a) Explain the method of wet suppression of dust.  
b) Explain the parallel operation of fans.
3. What are the desirable features of good mine ventilation planning.
4. Discuss the working of thermal precipitator for dust sampling.
5. a) Explain the working principle of Pitot static tube with neat sketch.  
b) Explain the different types of manometer use for pressure measurement with neat sketch.

MI-6002-CBGS

PTO

[2]

6. List out the different types of dust sampler use in mine with their working principle.
7. Describe various instruments used for measurement of pressure, velocity and quantity of air.
8. Explain in brief: (any four)
  - a) Sampling of Air Borne Dust
  - b) Evasee
  - c) Auxiliary fan
  - d) Losses in mine fan
  - e) Homotropical ventilation
  - f) Silicosis

\*\*\*\*\*